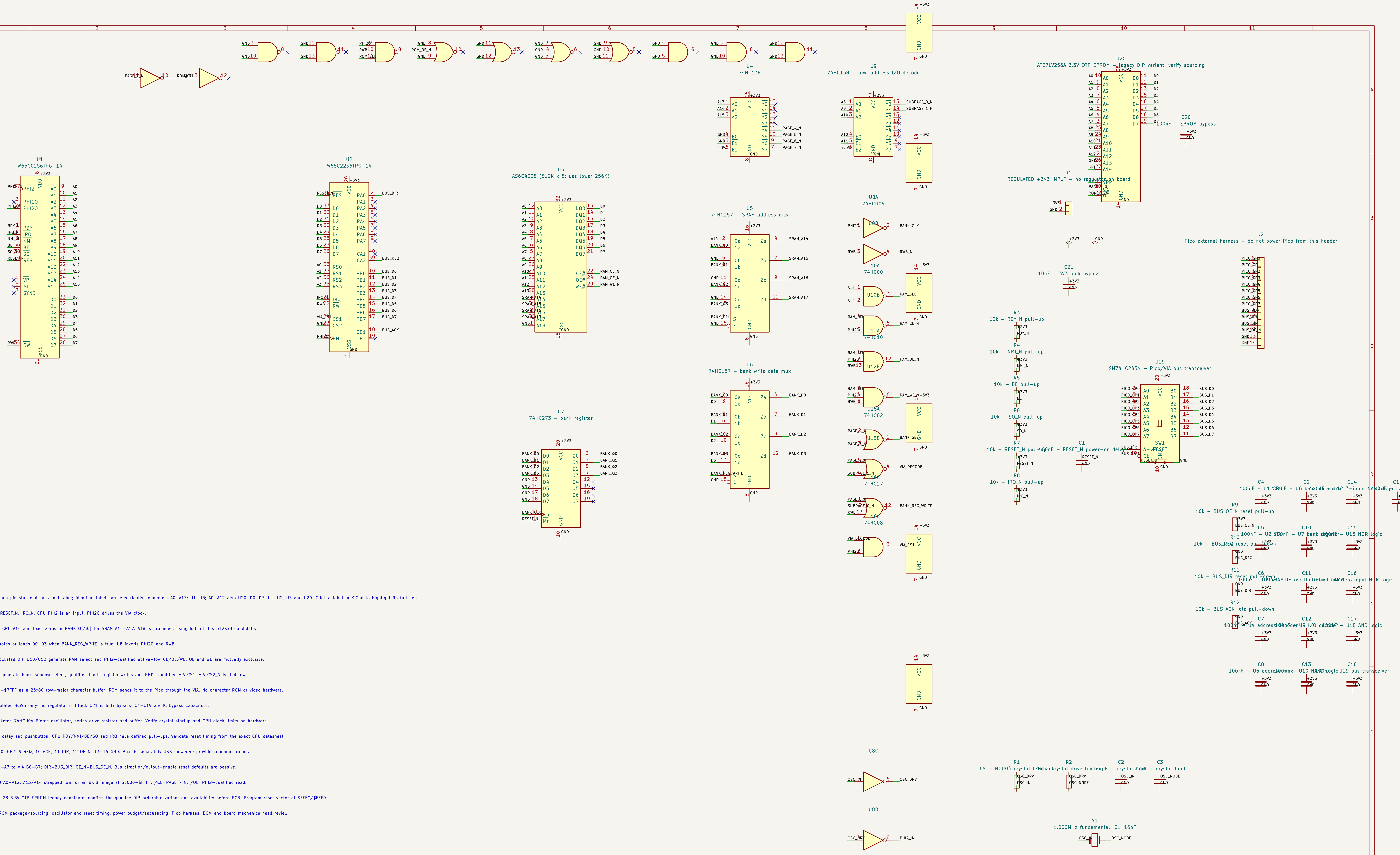
U1
W65C02S6TPG-14
U2
W65C22S6TPG-14
U3
AS6C4008 (512K x 8; use lower 256K)
U4
74HC138
U9
74HC138 – low-address I/O decode
U20
AT27LV256A 3.3V OTP EPROM – legacy DIP variant; verify sourcing
C20
100nF – EPROM bypass
J1
REGULATED +3V3 INPUT – no regulator on board
U5
74HC157 – SRAM address mux
U8A
74HCU04
U10A
74HC00
U6
74HC157 – bank write data mux
U15A
74HC02
U7
74HC273 – bank register
U16A
74HC27
U18A
74HC08
J2
Pico external harness – do not power Pico from this header
C21
10uF – 3V3 bulk bypass
R3
10k – RDY_N pull-up
R4
10k – NMI_N pull-up
R5
10k – BE pull-up
R6
10k – SO_N pull-up
R7
10k – RESET_N pull-up
C1
100nF – RESET_N power-on delay
SW1
RESET
U19
SN74HC245N – Pico/VIA bus transceiver
R8
10k – IRQ_N pull-up
R9
10k – BUS_OE_N reset pull-up
R10
10k – BUS_REQ reset pull-down
R11
10k – BUS_DIR reset pull-down
R12
10k – BUS_ACK idle pull-down
C4
100nF – U1 CPU
C5
100nF – U2 VIA
C6
100nF – U3 SRAM
C7
100nF – U4 address decoder
C8
100nF – U5 address mux
C9
100nF – U6 bank data mux
C10
100nF – U7 bank register
C11
100nF – U8 oscillator/inverter
C12
100nF – U9 I/O decoder
C13
100nF – U10 NAND logic
C14
100nF – U12 3-input NAND logic
C15
100nF – U15 NOR logic
C16
100nF – U16 3-input NOR logic
C17
100nF – U18 AND logic
C18
100nF – U19 bus transceiver
U8C
U8D
R1
1M – HCU04 crystal feedback
R2
crystal drive limiter
C2
27pF – crystal load
C3
27pF – crystal load
Y1
1.000MHz fundamental, CL=16pF
+3V3
GND
A0
D0
PHI2
PHI2O
RWB
RESET_N
IRQ_N
RDY_N
NMI_N
BE
SO_N
BUS_DIR
BUS_REQ
BUS_ACK
BUS_OE_N
BUS_D0
BANK_D0
BANK_Q0
BANK_SEL
BANK_CLK
BANK_REG_WRITE
SRAM_A14
SRAM_A15
SRAM_A16
SRAM_A17
RAM_SEL
RAM_CE_N
RAM_OE_N
RAM_WE_N
ROM_OE_N
PAGE_4_N
PAGE_5_N
PAGE_6_N
PAGE_7_N
SUBPAGE_0_N
SUBPAGE_1_N
VIA_DECODE
VIA_CS1
RWB_N
OSC_IN
OSC_DRV
OSC_NODE
PHI2_IN
PICO_GP0

HOW TO FOLLOW CONNECTIONS: each pin stub ends at a net label; identical labels are electrically connected. A0–A13: U1–U3; A0–A12 also U20. D0–D7: U1, U2, U3 and U20. Click a label in KiCad to highlight its full net.

CPU/VIA: A0–A15, D0–D7, RWB, RESET_N, IRQ_N. CPU PHI2 is an input; PHI2O drives the VIA clock.

SRAM: A0–A13 direct; U5 selects CPU A14 and fixed zeros or BANK_Q[3:0] for SRAM A14–A17. A18 is grounded, using half of this 512Kx8 candidate.

U7 resets the bank to zero; U6 holds or loads D0–D3 when BANK_REG_WRITE is true. U8 inverts PHI2O and RWB.

U4/U9 decode address pages. Socketed DIP U10/U12 generate RAM select and PHI2–qualified active–low CE/OE/WE; OE and WE are mutually exclusive.

Socketed DIP U15–U16 and U18 generate bank–window select, qualified bank–register writes and PHI2–qualified VIA CS1; VIA CS2_N is tied low.

Text screen: reserve SRAM $7800–$7FFF as a 25x80 row–major character buffer; ROM sends it to the Pico through the VIA. No character ROM or video hardware.

Power: J1 accepts externally regulated +3V3 only; no regulator is fitted. C21 is bulk bypass; C4–C19 are IC bypass capacitors.

Clock: 1.000MHz crystal with socketed 74HCU04 Pierce oscillator, series drive resistor and buffer. Verify crystal startup and CPU clock limits on hardware.

Reset: pull–up, 100nF power–on delay and pushbutton; CPU RDY/NMI/BE/SO and IRQ have defined pull–ups. Validate reset timing from the exact CPU datasheet.

Pico is external: J2 pins 1–8 GP0–GP7, 9 REQ, 10 ACK, 11 DIR, 12 OE_N, 13–14 GND. Pico is separately USB–powered; provide common ground.

U19 SN74HC245N routes Pico A0–A7 to VIA B0–B7; DIR=BUS_DIR, OE_N=BUS_OE_N. Bus direction/output–enable reset defaults are passive.

U20 AT27LV256A: A0–A12 to CPU A0–A12; A13/A14 strapped low for an 8KiB image at $E000–$FFFF. /CE=PAGE_7_N; /OE=PHI2–qualified read.

U20 is shown as a socketed DIP–28 3.3V OTP EPROM legacy candidate; confirm the genuine DIP orderable variant and availability before PCB. Program reset vector at $FFFC/$FFFD.

NOT fabrication–ready: exact EPROM package/sourcing, oscillator and reset timing, power budget/sequencing, Pico harness, BOM and board mechanics need review.

NOT PCB–ready: verify timing, power, bus interface, BOM and mechanics
Socketed–DIP 3.3 V design draft; legacy OTP EPROM sourcing TBD

Sheet: /
File: 6502–electrical–draft.kicad_sch

Title: 6502 Electrical Draft – CPU, VIA, SRAM and EPROM

Size: A2
Date: 2026–11–04
Rev: 0.2
KiCad E.D.A. 10.0.8
Id: 1/1